



Efficiency Incentives for A...
PDF

explain everything

Alright. Deep breath. This paper is dense, interdisciplinary, and ambitious. I'll explain **everything**—not just what it says, but *why it says it, how the pieces fit, and what you should actually understand and remember*.

No shortcuts, no hand-waving.

Everything below is derived from “AI Cap-and-Trade: Efficiency Incentives for Accessibility and Sustainability” (Bornstein & Bedi, 2026)  Efficiency Incentives for Acces...

0. One-Sentence Summary (Anchor This First)

The paper argues that AI's current “bigger-is-better” race is economically exclusionary and environmentally harmful, and proposes a market-based cap-and-trade system on AI inference FLOPs to force efficiency, lower emissions, and let smaller players compete.

If that sentence makes sense by the end, you've succeeded.

1. The Core Diagnosis: What Is Going Wrong with AI Today?

1.1 Hyper-scaling is the dominant strategy

AI progress today follows a simple rule:

- Bigger models
- More data
- More GPUs
- More FLOPs (floating-point operations)

This works, but it's **brute force**.

Why companies do it:

- Scaling is predictable
- Efficiency is hard
- Investors reward performance, not elegance

1.2 FLOPs are the real currency of AI

Training and inference cost is measured in FLOPs.

Concrete numbers (from the paper):

- Frontier training $\approx 10^{27}$ FLOPs (1 ronnaFLOP)
- A single LLM query $\approx 8 \times 10^{14}$ FLOPs
- ~2.5 billion prompts/day $\Rightarrow$ ~1 ronnaFLOP/year just for inference

Inference, not training, is the long-term killer.

2. Two Systemic Failures Caused by This Model

2.1 Accessibility failure (economic)

What's happening

- GPUs are scarce and expensive
- Compute is centralized
- Only a few firms can deploy at scale

Results:

- Academics can't compete
- Startups can't deploy
- Research talent drains into big tech
- AI becomes an **oligopoly**

The paper explicitly frames this as **barriers to entry**, not lack of ideas.

2.2 Sustainability failure (environmental)

Energy reality

- Every FLOP consumes electricity
- Electricity is mostly non-renewable

- Cooling + water usage explode with scale

Concrete emissions example:

- ~0.3 Wh per LLM query
- At scale: **hundreds of tons of CO₂ per day**
- EPA defines 100 tons/year as a *major polluter*

AI firms already exceed that by orders of magnitude.

Key insight:

Even if emissions per query go down, **total emissions rise because usage explodes.**

Efficiency gains are overwhelmed by scale.

3. Why Regulation Alone Is Not Enough

The authors make a critical governance argument:

- Traditional AI governance = **rules and restrictions**
- But regulation:
 - Slows innovation
 - Hits startups hardest
 - Encourages regulatory arbitrage (moving locations)

What's missing:

Market incentives that reward efficiency instead of punishing scale blindly

This is where economics enters.

4. Market-Based Incentives: The Toolbox

Before proposing anything new, the paper reviews known tools.

4.1 Charge systems (penalize inefficiency)

- **Pigouvian taxes:** tax negative externalities (e.g., carbon)
- **User fees:** token taxes, usage taxes
- **Credits & subsidies:** reward efficient or clean behavior
- **Deposit-refund systems:** pay upfront, get refunded if behavior is good

Problem:

- Taxes can be avoided
 - Subsidies distort markets
 - Fees don't directly shape model design
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4.2 Tradable permit systems (control quantity, not price)

This is the important part.

Two types:

1. **Credit programs** – reward being below baseline
2. **Cap-and-trade** – enforce a hard cap and allow trading

The authors argue:

Cap-and-trade is superior because total usage is capped, not just incentivized downward.

5. Cap-and-Trade (Explained From First Principles)

This section is the backbone of the paper.

5.1 What cap-and-trade means (in general)

- A regulator sets a **cap** on total usage (pollution, emissions, etc.)
- Companies receive **allowances**
- Allowances can be **traded**
- Total usage cannot exceed the cap

This is how carbon markets work.

5.2 Translating this to AI

Replace:

- CO₂ → **FLOPs**
- Carbon allowances → **AI Allowances**
- Pollution → **Inference computation**

Important design choice:

Only inference is capped, not training

Why?

- Training drives research
- Inference drives revenue and emissions
- Inference scales with users

This avoids killing innovation.

6. AI Allowances: How They Are Distributed

This is where fairness enters.

6.1 Why equal distribution fails

Giving everyone the same FLOP budget would:

- Cripple large providers
- Force user caps
- Cause "AI leakage" (companies relocate)

So they borrow from emissions trading systems.

6.2 Benchmarking (the key mechanism)

Each company gets allowances based on:

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$$A_i = O_i \times B \times C_i$$

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Where:

- O_i = output level (FLOPs historically used)
- B = efficiency benchmark (watts per FLOP)
- C_i = assistance factor

Interpretation:

- Efficient companies get surplus
- Inefficient companies must buy allowances
- Benchmark tightens over time

This **rewards efficiency directly**.

6.3 What companies can do with allowances

- Use them for inference
- Sell excess allowances
- Bank them for future years

This creates:

A real market where efficiency has monetary value

Small, efficient companies gain leverage.

7. The Economic Model (Yes, the Math Matters)

The paper doesn't just hand-wave—it proves claims.

7.1 Performance vs FLOPs

Assumption:

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$$\text{Loss} \propto 1 / x^k \quad (k > 0)$$

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Meaning:

- More FLOPs → better performance
- But diminishing returns

7.2 Cost of FLOPs

Each FLOP has a linear cost:

ini

$$\text{Cost} = a \times x$$

 Copy code

Electricity, cooling, infrastructure.

7.3 Utility without governance

Companies maximize:

$$u(x) = -x^{-k} - a \cdot x$$

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Solution:

$$x^* = (k / a)^{1/(k+1)}$$

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This is the **natural equilibrium** today.

7.4 Utility with cap-and-trade

Now add:

- FLOP limit F_i
- Market price per FLOP b
- Buying/selling FLOPs (y)

New utility:

$$u(x, y) = -x^{-k} - a \cdot x + b \cdot y$$

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Constraint:

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$$x + y \leq F_i$$

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Solution:

$$x^* = (k / (a + b))^{1/(k+1)}$$

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Critical result

Because $(a + b) > a$, it follows that:

$$x \text{ with cap-and-trade} < x \text{ without governance}^{**}$$

This is **provable FLOP reduction**.

8. Even Better: Utility Can Increase

Counterintuitive result:

- Under reasonable conditions
- With enough allowances

Companies can:

- Use fewer FLOPs
- Pay less for compute
- Earn revenue by selling allowances

So:

Efficiency can increase profits, not reduce them

This is the Porter Hypothesis applied to AI.

9. Why This Helps Small Players

This is not theoretical fluff.

Small companies:

- Often build more efficient systems
- Can't scale brute force
- Can sell unused allowances

Result:

- New revenue stream
- Survival mechanism
- Competitive balance

This directly addresses accessibility.

10. Environmental Impact

Because inference dominates emissions:

- Capping inference FLOPs
- Automatically caps energy use
- Automatically caps emissions

No need to track carbon explicitly.

Efficiency → sustainability emerges naturally.

11. Addressing the Objections

The authors take critics seriously.

Objection 1: Regulation kills innovation

Response:

- Cap-and-trade preserves freedom
- Doesn't dictate how models are built
- Only prices inefficiency

Objection 2: National security requires scaling

Response:

- Training is uncapped
- Dominance isn't prevented
- Wasteful inference is discouraged

Objection 3: Startups can't handle compliance

≡ ChatGPT 5.2 ▾

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- Allowances are free
 - Benchmarked, not uniform
 - Startups benefit most from efficiency markets
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12. Final Intellectual Takeaways (Memorize These)

1. AI's biggest externality is inference FLOPs
 2. Efficiency is undervalued because scale is cheaper to optimize
 3. Market incentives shape behavior better than rules
 4. Cap-and-trade controls quantity, not innovation
 5. Efficiency can be monetized
 6. Accessibility and sustainability are linked, not separate
 7. Inference governance is safer than training governance
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13. What This Paper Really Is

Not:

- A regulation proposal
- An environmental manifesto
- An anti-AI argument

It is:

A mechanism design paper for the AI economy.

It asks:

“What if efficiency, not scale, paid the bills?”

If you want, next we can:

- Map this directly to **LLM deployment economics**
- Compare **cap-and-trade vs token taxes**
- Discuss **whether this could realistically be enforced**
- Or translate this into **exam-ready notes or a presentation**

But now, you have **everything**.

Sources

